Presenter's Guide – Easwari ERP Portal

What to say on each slide, how to explain the technology stack, and likely viva questions · OOPJ (2321CSC304R)

Part 1 – How to explain the stack (the 30-second version)

> *"Our system has three parts. The browser part is React, which shows the Student, Faculty and Admin screens. The server part is Java 17 with Spring Boot, which holds all the rules, checks who you are with a JWT token, and saves data through JPA into an H2 database. A small Python FastAPI service calculates the risk score. The browser talks to the server in JSON over HTTPS; the server talks to the database and to the scoring service."*

Technology	How to explain it
Java 17	The main programming language of the back end. This is an OOPJ project, so mention classes, inheritance of repositories, interfaces and exceptions.
Spring Boot 3.2.5	Framework that gives a ready web server, REST controllers, dependency injection and validation, so we write business rules instead of plumbing.
Spring Data JPA / Hibernate	Maps Java objects (entities) to database tables, so no hand-written SQL for most operations. Our unique constraint is declared on the entity.
H2 database → PostgreSQL	H2 is an embedded file database needing no installation but still enforcing constraints. For production only the connection settings change to PostgreSQL.
Spring Security, JWT, BCrypt	Spring Security guards the routes by role. JWT is the signed token the browser sends with each request. BCrypt hashes passwords, so they are never stored in plain text.
React 18 + Vite 5	React builds the screens from reusable components; Vite is the fast build tool. Recharts draws the charts and pdf.js reads resume PDFs in the browser.
FastAPI (Python) + NumPy	A small separate service that calculates the risk score and placement-readiness score. It has no database; Java sends it the numbers and gets a score back.
Apache POI 5.2.5	Java library to read Excel files – used for bulk account import and exports.
Maven	Build tool that compiles, tests and packs the project into one executable jar (about 72 MB).

JUnit 5	Unit-testing library; 11 tests cover the attendance, marks, faculty and password rules.

Part 2 – Slide-by-slide script

Slide 1 – Title

“Good morning. We are Rehan, Sakthisaran and Sanjay from II Year CSE, Section E. Our project for Object Oriented Programming using Java is the Easwari ERP Portal, an Intelligent Attendance Control and Student Progress Monitoring Platform.”

Tip: Say the project name, team and subject once. Keep it under 20 seconds.

Slide 2 – Introduction

“In a college, attendance decides whether a student can write the exam, because of the 75% rule, and internal marks count towards the grade. Today these live in paper registers and spreadsheets. Our portal brings attendance, marks, results, fees, leave and placement into one web platform with three portals – Student, Faculty and Administrator – behind one sign-in. The core idea is: store each fact once, and let the server decide who may change it.”

Tip: Stress the last sentence – it is the heart of the project.

Slide 3 – Problem Statement

“There are four problems. One: no single record, so the same student shows different attendance on different lists. Two: no control over authority – any teacher can edit any class and corrections leave no trail. Three: warning comes late – short attendance is noticed only when the eligibility list is published. Four: services are scattered – fees, leave and placement are at different counters.”

Slide 4 – Objectives

“Our objectives were: three role-based portals with one secure sign-in; attendance per period with a database constraint so it cannot be marked twice; the class coordinator controls the whole class while a subject teacher controls only their own subject; marks visible to students only after publishing; fees, leave, services, placement and a Career Hub; and an explainable risk score.”

Slide 5 – Literature Survey

“We studied three papers. Ramadhani and Satria built a QR-code attendance system, but it does not say who may change a record. Akçapınar and colleagues predict at-risk students from online-learning logs, but that needs an LMS. Howard and colleagues compared eight prediction methods and found continuous assessment marks are the best predictor – this is why internal marks carry weight in our risk score. The gap: nobody defines authority over records for an affiliated engineering college.”

Slide 6 – Proposed System & Architecture

“This is the layered architecture. At the top is the browser, running React, with the three portals. It talks over HTTPS using JSON and a bearer token to the application layer, which is Spring Boot in Java 17. That layer has a security filter, controllers, services, validation and an audit service. Below it is the data layer – an H2 file database for development and PostgreSQL for production, changed only by configuration. On the side is a separate FastAPI scoring service. It has no database connection, so the portals still work if it is stopped.”

Tip: Point at the diagram from top to bottom while you speak.

Slide 7 – Technology Stack & Modules

“Explain the table row by row – see the stack guide in the next section. Then name the six modules: authentication and accounts, attendance, marks and examinations, fees services and leave, administration and audit, and the risk score with the Career Hub.”

Slide 8 – Attendance Authority & Data Model

“Every attendance request passes a check. Is the user faculty? Do they teach this offering, or coordinate this class? Are the students in this class? If any answer is no, the server refuses with 403 or 400. If all yes, rows are saved as an upsert. Attendance is stored once per student, offering, date and period under the unique constraint uk_attendance_slot, so marking twice only updates the row. The data model on the right is centred on the subject offering – one subject, taught to one section by one teacher – and attendance and assessments hang from it.”

Slide 9 – Security & Sign-in

“Sign-in returns a signed JWT with the role. New accounts get a random 10-character temporary password that must be changed at first login. Students can then set a 6-digit quick code, which locks for 15 minutes after 5 wrong attempts. The token is kept in memory only, so nothing remains on a shared lab computer. Security has three layers: route guards on /api/admin, /api/faculty and /api/student; role annotations on controllers; and ownership checks in the services. Every privileged admin action is written to an audit trail.”

Slides 10–12 – Student, Faculty, Admin portals

“Walk through the screenshots. Student: dashboard with CGPA, attendance and fees, and the attendance register grid. Faculty: the coordinator can mark every period of the class; a subject teacher sees other periods as view-only; marks are entered per assessment as a draft and published later. Admin: dashboard, risk and alerts page, and the Career Hub tools such as the resume analyzer.”

Tip: Note: screens show a seeded trial data set, not real college data. Say so if asked.

Slide 13 – Testing & Results

“We ran 68 REST test cases on the packaged application – 67 behaved as expected and one revealed a defect – and all 11 JUnit unit tests passed. A subject teacher was refused with 403 for a subject that is not theirs, and re-marking a period updated the row without changing the percentage. The risk score table shows a healthy student scores 0 (LOW), a borderline one 36 (MEDIUM) and a struggling one 66 (HIGH). The one defect: an unknown status value on the service-request route returns HTTP 500 instead of 400.”

Tip: Being open about the defect looks honest and strong.

Slide 14 – Conclusion, Limitations, Future Scope

“In conclusion, each fact is stored once and each rule is enforced on the server. Limitations: payment is simulated, the data is a seeded trial set, there are two known defects, and the risk weights need calibration. Future scope: a real payment gateway, real college data, alerts near 75%, two-factor login, QR or mobile attendance, and parent, mentor, hostel and library modules.”

Slide 15 – Thank You

“Thank you. We are happy to take questions.”

Part 3 – Questions you may be asked

Q. Why Spring Boot?

A. It removes boilerplate and gives security, validation and REST support out of the box, so we could focus on the rules of attendance and marks.

Q. Why is attendance stored per period?

A. It is the finest grain. Day and subject views are calculated from the same rows, so they can never disagree.

Q. What stops a period being marked twice?

A. The unique constraint uk_attendance_slot on student, offering, date and period; the service updates the existing row.

Q. Who can change attendance?
A. The administrator; the class coordinator for every subject of the class; and a teacher only for his or her own subject.

Q. Why is the student's identity taken from the token?
A. So changing a number in the address cannot expose another student's record.

Q. Why a separate scoring service?
A. Java owns the data and Python owns the scoring. Each can be scaled or replaced independently, and the portals work even when scoring is off.

Q. Is the risk score a prediction?
A. No. It is a transparent weighted score – attendance 0.35, CGPA 0.30, trend 0.20, eligibility 0.15 – that lists its factors. The weights still need calibration with real outcomes.

Q. How is a password protected?
A. Temporary passwords are random, shown once, stored hashed with BCrypt and must be changed at first login.

Q. Is the payment real?
A. No, it is simulated; no card or bank details are collected.

Q. Is the data real?
A. No. It is a seeded trial data set of 5 departments, 10 classes, 59 students, 25 staff and 28,497 attendance records.

Q. What are the known limitations?
A. Simulated payment, trial data, the unknown-status 500 error, the reset-password default, and development defaults for the signing secret and H2 console.

Part 4 – Delivery tips

- Roughly 1 minute per content slide; spend extra time on slides 6, 8 and 13.
- Split the speaking among the three members – for example one takes slides 2–5, one takes 6–9, one takes 10–14.
- If you do not know an answer, say what the system does and avoid guessing numbers.
- Open the live portal if the examiner asks for a demo; the screenshots are from a seeded trial data set.